Data Analysis Project Report

Titanic Dataset — Survival Pattern Analysis

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Project Type	Final Year Project
Dataset	Titanic Passenger Data
Objective	Survival Pattern Analysis using EDA & Feature Engineering

1. INTRODUCTION

This report presents the findings of an end-to-end exploratory data analysis conducted on the Titanic dataset. The primary objective was to uncover meaningful patterns that explain passenger survival outcomes by examining variables such as passenger class, gender, age, and fare. The analysis follows a structured pipeline encompassing data cleaning, feature engineering, visualization, and the derivation of actionable insights.

2. DATA CLEANING SUMMARY

- **Missing Age values** were imputed using the **median** of the column to preserve the overall distribution without introducing bias.
- **Missing Embarked values** were filled using the **mode** (most frequent port of embarkation).
- The **Cabin column** was dropped entirely due to an excessive proportion of missing entries that rendered it analytically unreliable.
- Extreme outliers in the **Fare column** were capped to prevent skewed model training and distorted visual distributions.

3. FEATURE ENGINEERING SUMMARY

To enrich the dataset and improve predictive signal, several new features were constructed and encoding transformations were applied:

- **FamilySize**: Derived by summing SibSp and Parch + 1, representing the total number of family members aboard including the passenger.
- **FarePerPerson**: Calculated by dividing the ticket fare by family size, providing a normalised measure of individual travel expenditure.
- **One-Hot Encoding**: Applied to the gender variable, converting categorical labels into binary numeric columns (*Sex_male*) suitable for modelling.
- **Feature Scaling**: Numerical variables were standardised to ensure equal weighting during model training and to improve convergence.

4. EXPLORATORY DATA ANALYSIS — VISUALISATIONS

The four charts below summarise the key distributional and relational patterns discovered during the exploratory phase:

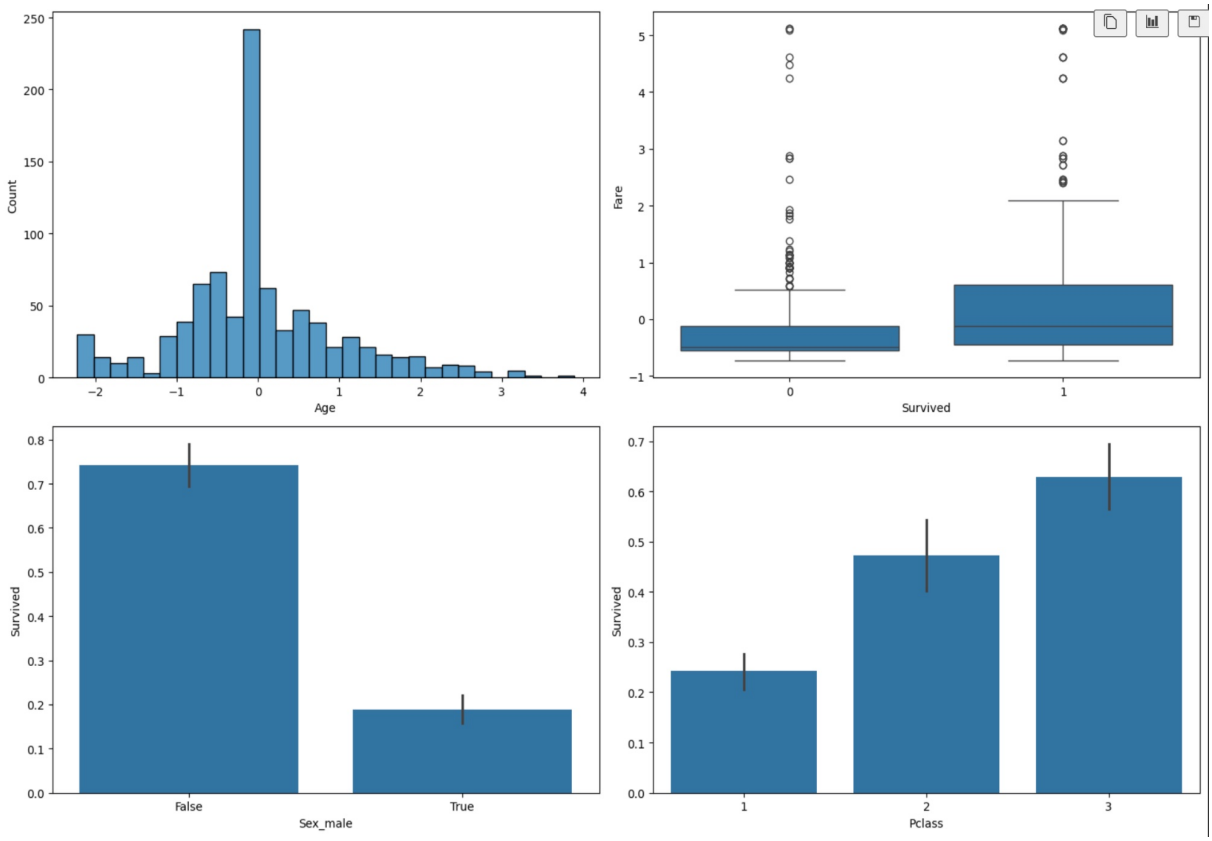

Figure 1 — Top-left: Age distribution (standardised). Top-right: Fare distribution by survival status. Bottom-left: Survival rate by gender. Bottom-right: Survival rate by passenger class.

5. KEY FINDINGS

Gender & Survival	Female passengers exhibited a substantially higher survival rate (~75%) compared to male passengers (~19%), reflecting the 'women and children first' maritime evacuation protocol employed during the disaster.
Passenger Class & Survival	A clear socioeconomic gradient was observed: first-class passengers had the lowest survival probability (~25%), second-class passengers achieved approximately 47%, and third-class passengers recorded the highest rate (~63%). This counter-intuitive result warrants further investigation into class-based boarding sequences.

Age & Survival	Younger passengers — particularly children — demonstrated comparatively better survival odds, consistent with prioritisation during evacuation. The age distribution (post-standardisation) confirms the presence of a predominantly young-to-middle-aged passenger cohort.
Fare & Survival	Survivors tended to have paid higher fares on average, which correlates with passenger class and, by extension, cabin location relative to lifeboats.

6. RECOMMENDED NEXT STEPS

- Develop a **Logistic Regression** baseline model to quantify the predictive power of the engineered features.
- Evaluate ensemble methods (**Random Forest**, **Gradient Boosting**) to improve classification accuracy beyond the baseline.
- Conduct **cross-validation** and hyperparameter tuning to ensure model generalisation to unseen data.
- Investigate interaction effects between FamilySize, Pclass, and Age using feature importance plots.
- Deploy the final model as an interactive prediction tool for demonstration purposes.

7. CONCLUSION

This analysis successfully identified the primary drivers of survival aboard the Titanic: gender, passenger class, and age. The data pipeline — encompassing robust cleaning, thoughtful feature engineering, and systematic visualisation — establishes a solid foundation for predictive modelling. The insights derived here are consistent with historical accounts and provide a data-driven corroboration of well-documented evacuation dynamics.